

Aarya Thakur

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SUMMARY

Final year AI DS student with hands-on experience building LLM-powered systems, agentic AI pipelines, and RAG architectures. Proven in high-pressure environments with 2 internships and 6 hackathon wins.

EDUCATION

Thadomal Shahani Engineering College CGPA 9.43
Bachelor of Engineering in Artificial Intelligence and Data Science November 2022 – June 2026

EXPERIENCE

Fullstack Intern June 2025 – July 2025
Augenblick Consulting Remote

- Implemented **Deep Zoom Technology**, improving large image rendering speed and enabling seamless web-based navigation of **gigapixel images**.
- Contributed to building a **RAG-powered system** that converts RIS files into **vector embeddings**, enabling accurate research paper classification, tagging, and advanced rule-based filtering.

AI Intern January 2025 – April 2025
Nimrobo AI Remote

- Created a tool enabling **agentic AI** to query and interpret CSV data using **SQL** for advanced data analysis and insights.
- Integrated **citation tracking** into AI Agent outputs, enhancing content credibility and reducing **hallucinated outputs**.
- Developed a **container orchestration system** that automatically spins up isolated AI agent instances with **WebSocket** access, and automated lifecycle management, improving scalability and reliability.

PROJECTS

MatryoshkaSearch [↗](#) | *Retrieval Pipeline with Custom Embeddings* | Python, LangGraph, FastAPI, ChromaDB

- Fine-tuned embedding model on ~7K financial Q&A pairs using **Matryoshka Representation Learning** achieving **10–13% gains** on various metrics over the base model.
- Built a **LangGraph** retrieval pipeline with **semantic chunking**, query routing and **cross-encoder reranking**, containerized the full stack with Docker.

CollabNext [↗](#) | *Real-Time Collaborative Editor with AI* | Next.js, TypeScript, Firebase, WebSocket, Yjs

- Built a real-time collaborative content studio with **WebSockets** and **Yjs CRDTs**, ensuring conflict-free synchronization and smooth multi-user editing.
- Implemented a **Simulated Annealing**-based solution, enabling escape from local optima and consistently outperforming greedy baseline heuristics.

Metaheuristic Job Scheduler [↗](#) | *Simulated Annealing & Genetic Algorithms* | Python

- Implemented a **Simulated Annealing**-based solution, enabling escape from local optima and consistently outperforming greedy baseline heuristics.
- Applied **Genetic Algorithm** with custom chromosome encoding, crossover, and mutation strategies, achieving faster convergence and superior global exploration.

ACHIEVEMENTS

Winner of **6 Hackathons**, consistently excelling in fast-paced, high-stakes environments across diverse technical domains.

Achieved a **Perfect 10 CGPA** in the 6th Semester.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL

Databases: PostgreSQL, MySQL, Firebase, Vector Databases

Frameworks/Libraries: Next.js, Django, Node.js, Express.js, Tailwind CSS, NumPy, Pandas, PyTorch, LangGraph

Tools/Platforms: Git, GitHub, Docker, Linux, Vercel